06 - Dataset and Data Preprocessing

What data exists and how it is prepared

Knowledge-pack rule: every statement below is derived from files in this project or is explicitly labelled as planned, missing or not verified. Nothing here is an invented citation, dataset, result or feature.

Source files used for this document

documentation/dataset_description.md; src/data_collection/synthetic.py; src/preprocessing/*.py; src/feature_engineering/features.py; survey/survey_data.csv

Datasets

Dataset	Origin	Notes
Transactions	Seeded synthetic generator (src/data_collection/synthetic.py)	Fraud ~2%; four stylised patterns each expressed in 75% of its cases; ~0.4% injected defects
Security events	Seeded synthetic generator	18% attacks; six campaign bursts; benign background
Survey	Synthetic latent-factor generator; saved to survey/survey_data.csv	Placeholder for pipeline demonstration - NOT real respondents
ULB credit-card fraud	Optional public dataset (user downloads to data/external)	Code path exists (src/data_collection/external.py); no results exist unless the file is supplied

All numbers in this pack derived from these datasets describe the method on synthetic data. They are not evidence about real-world fraud prevalence or real consumers.

Transaction columns

Column	Meaning
transaction_id, customer_id	surrogate keys
timestamp	90-day window anchored to the run date when end_date is auto
amount	log-normal around a per-customer level
merchant_category, channel, device_type	categoricals (11, 4, 5 values)
country, home_country	used to derive is_foreign
is_new_device, ip_risk_score, velocity_1h	risk signals
account_age_days, auth_method	tenure and authentication
is_fraud	label
fraud_pattern	generator pattern; analysis only, never a feature

Fraud patterns in the generator

Pattern	Share of fraud	Signature
account_takeover	40%	new device, foreign, high IP risk, amount x2-6, night
card_testing	25%	tiny amounts, high velocity, web, digital goods
synthetic_identity	15%	very new account, amount x3-8, new device

Pattern	Share of fraud	Signature
social_engineering	20%	normal device and geography, large OTP-authorised transfer

Actual data volumes from the last pipeline run

Quantity	Value
Run id	run_20261109_061947_57b9
Rows in (raw)	60120
Duplicates removed	120
Invalid rows removed	318
Rows out (clean)	59682
Fraud rate	2.0207%
Validation pass rate	0.769

Validation results

Check	Passed	Failed rows
schema_columns_present	True	0
min_rows	True	0
missing_ratio_within_limit	True	0
no_missing_amount	False	242
duplicate_ratio_within_limit	True	0
no_duplicate_transaction_id	False	120
amount_positive	False	78
timestamp_parsable	True	0
ip_risk_in_unit_interval	True	0
label_is_binary	True	0
channel_domain	True	0
merchant_domain	True	0
auth_method_domain	True	0

Failed checks on the raw data are expected: the generator deliberately injects missing amounts, negative amounts and duplicate rows so validation and cleaning have real work.

Validation and cleaning

- src/preprocessing/validation.py: 13 checks (schema columns, minimum rows, missing ratio, missing amount, duplicate ratio and ids, positive amounts, parsable timestamps, IP risk within 0-1, binary label, channel/merchant/auth domains).
- src/preprocessing/cleaning.py: drop duplicate transaction ids, drop missing/non-positive amounts and unparsable timestamps, clip IP risk, fill categorical gaps with 'unknown', sort by time, return a row-accounting report; frame_hash for provenance.

Feature engineering

src/feature_engineering/features.py builds 17 numeric and 23 one-hot features (40 total).

Group	Features
Time	hour, day_of_week, is_night, is_weekend
Amount	log_amount, amount_to_cust_avg_ratio
Customer history (past only)	cust_txn_count_prior, log_secs_since_last, amount ratio
Risk flags	is_new_device, is_foreign, ip_risk_score, velocity_1h, high_risk_ip, is_new_account, log_account_age
Interactions	new_device_x_foreign, night_x_high_ratio
One-hot	channel, merchant_category, auth_method, device_type

Leakage protection: history features use cumulative sums minus the current row, so a transaction never sees its own or future values. tests/test_preprocessing.py includes a test that changes the last row and checks earlier features are unchanged.

Survey privacy handling

- respondent_id is a 12-character salted SHA-256 prefix (salt from SURVEY_SALT).
- Ages are bands; no free text, IP or email fields.
- k-anonymity check and suppression on age_band x education x usage_frequency (k=5).

Storage

Parquet and CSV in data/raw, data/synthetic, data/interim, data/processed. These folders are git-ignored and regenerated; only the survey CSV is tracked. Files currently present:

```
data/README.md
data/external/.gitkeep
data/interim/.gitkeep
data/interim/transactions_clean.csv
data/interim/transactions_clean.parquet
data/processed/.gitkeep
data/processed/security_events_scored.parquet
data/processed/transactions_features.parquet
data/processed/trust_scores.csv
data/processed/trust_scores.parquet
data/raw/.gitkeep
data/raw/security_events_raw.csv
data/raw/transactions_raw.csv
data/synthetic/.gitkeep
data/synthetic/security_events_synthetic.parquet
data/synthetic/transactions_synthetic.parquet
```